

YouTube Data API v3

Usage Script & End Results

Project BTC News Shorts & AI Music Publishing

GCP Project btc-news-shorts

Date 2026-03-19

Channels 4 YouTube channels (3 News + 1 Music)

1. Project Overview

We operate two automated pipelines under one GCP project, serving a total of 4 YouTube channels:

Pipeline A — BTC News Shorts

- Fetches the latest Bitcoin/crypto news from public APIs
- Generates narration scripts and TTS audio
- Composes vertical short-form videos (1080×1920, <60s)
- Uploads to YouTube via Data API v3 across 3 language channels (EN, KO, JA)
- Attaches SRT caption files for accessibility and SEO
- Publishes 1–2 videos per channel per day

Pipeline B — AI Music Publishing

- Generates original music tracks via AI composition
- Combines audio with cover art into video format (MP4)
- Uploads to YouTube via Data API v3 (1 music channel)
- Publishes 1–3 tracks per day

2. YouTube API Endpoints Used

Endpoint	Purpose	Quota Cost	Used By
videos.insert	Upload MP4 video	1,600 units	Both
captions.insert	Attach SRT subtitle	400 units	News only

Quota Breakdown Per Run

Pipeline	Per Upload	Uploads/Run	Cost/Run
News (3 langs)	2,000 units	3	6,000 units
Music	1,600 units	1–3	1,600–4,800 units
Combined daily	—	—	7,600–16,800 units

3. Authentication Flow

- OAuth 2.0 with offline refresh tokens
- Scopes: youtube.upload, youtube.force-ssl
- Separate token files per channel:
 - youtube_token_en.json (News EN)

- youtube_token_ko.json (News KO)
- youtube_token_ja.json (News JA)
- youtube_token.json (Music)
- First-time auth via browser-based InstalledAppFlow, then cached & auto-refreshed

4. Upload Source Code

4-A. News Pipeline — youtube_uploader.py

```
python

def upload_video(video_path, title, description, tags, privacy, token_filename):
    """Upload a video to YouTube and return the video ID."""
    youtube = get_authenticated_service(token_filename)

    body = {
        "snippet": {
            "title": title,
            "description": description,
            "tags": tags,
            "categoryId": "25", # News & Politics
        },
        "status": {
            "privacyStatus": privacy,
            "selfDeclaredMadeForKids": False,
        },
    }

    media = MediaFileUpload(
        str(video_path), mimetype="video/mp4",
        resumable=True, chunksize=1024 * 1024,
    )

    request = youtube.videos().insert(
        part="snippet,status", body=body, media_body=media,
    )

    response = None
    while response is None:
        status, response = request.next_chunk()

    return response["id"]
```

```
python

def upload_captions(video_id, srt_path, language, token_filename):
    """Upload an SRT caption track to a YouTube video."""
    youtube = get_authenticated_service(token_filename)

    body = {
        "snippet": {
            "videoId": video_id, "language": language,
            "name": "", "isDraft": False,
        },
    }

    media = MediaFileUpload(str(srt_path), mimetype="application/x-subrip")
    youtube.captions().insert(
        part="snippet", body=body, media_body=media,
    ).execute()
```

4-B. Music Pipeline — youtube_uploader.py

```
python

def upload_audio_as_video(audio_path, cover_path, title, description, tags):
    """Combine audio + cover image into MP4 via FFmpeg, then upload."""
    video_path = _create_video_from_audio(audio_path, cover_path)
    youtube = get_authenticated_service()

    body = {
        "snippet": {
            "title": title, "description": description,
            "tags": tags, "categoryId": "10", # Music
        },
        "status": {
            "privacyStatus": "public",
            "selfDeclaredMadeForKids": False,
        },
    }

    media = MediaFileUpload(
        str(video_path), mimetype="video/mp4",
        resumable=True, chunksize=1024 * 1024,
    )

    request = youtube.videos().insert(
        part="snippet,status", body=body, media_body=media,
    )

    response = None
    while response is None:
        status, response = request.next_chunk()
    return response["id"]
```

5. Pipeline Execution Scripts (CLI)

```
bash

# News: Generate videos for all 3 languages and upload
python -m src.pipeline --all-langs --upload

# Music: Generate and upload a track
python main.py --upload
```

6. Actual Execution Log (2026-03-19)

Below is the real terminal output from today's News pipeline run:

```

=====
BTC News Pipeline - 20260319_072039
Languages: EN, KO, JA
=====

[Quota] 6,000/10,000 used today (2026-03-18 PT)
Remaining: 4,000 units | Can upload: 2 more videos

[QUOTA] Can only upload 2/3 languages. Will upload first 2.

[1] Fetching Bitcoin news...
-> 36 articles found
-> BTC: $71,108 (-4.56%) | Sentiment: bearish

[2] Generating English narration script... -> 142 words
[3] Generating title & description...
-> Title: Fed Freezes Rates, BTC Dumps to $71,100

[4] Generating videos for 2 language(s)...

--- [EN] English ---
[OK] Audio: btc_20260319_072039_en.mp3 (245.1 KB)
[OK] Video: btc_20260319_072039_en.mp4 (16.5 MB, 62.7s)
Uploading to YouTube (English)...
-> 6% ... 96% uploaded
[OK] Upload complete: https://youtube.com/shorts/H0eosxyQ56w
[OK] Caption uploaded: en -> btc_20260319_072039_en.srt

--- [KO] Korean ---
[OK] Audio: btc_20260319_072039_ko.mp3 (228.8 KB)
[OK] Video: btc_20260319_072039_ko.mp4 (16.6 MB, 58.6s)
Uploading to YouTube (Korean)...
-> 6% ... 96% uploaded
[OK] Upload complete: https://youtube.com/shorts/hTdSES36tk8
[OK] Caption uploaded: ko -> btc_20260319_072039_ko.srt

--- [JA] Japanese ---
[SKIPPED] Insufficient daily quota

=====
Pipeline complete!
[EN] -> https://youtube.com/shorts/H0eosxyQ56w
[KO] -> https://youtube.com/shorts/hTdSES36tk8
[JA] -> Skipped (quota exceeded)
=====

```

7. End Results

Uploaded Videos (News — 2026-03-19)

Language	Title	YouTube URL
English	Fed Freezes Rates, BTC Dumps to \$71,100	youtube.com/shorts/H0eosxyQ56w
Korean	연준 금리 동결, BTC \$71,100로 급락	youtube.com/shorts/hTdSES36tk8
Japanese	(skipped — quota exceeded)	—

Video Specifications

Spec	News Shorts	Music
Resolution	1080×1920 (9:16)	1920×1080
Duration	~60 seconds	2–5 minutes
Format	MP4 (H.264 + AAC)	MP4 (H.264 + AAC 192k)
Category	News & Politics (25)	Music (10)
Captions	SRT per language	N/A

8. Why We Need a Quota Increase

Current Situation

Pipeline	Daily Quota Need	Channels
News (3 langs × 1–2 runs)	6,000–12,000 units	3
Music (1–3 tracks)	1,600–4,800 units	1
Combined daily total	7,600–16,800 units	4
Current daily limit	10,000 units	—

The Problem

The default 10,000-unit quota is insufficient for both pipelines operating under the same GCP project. As demonstrated in today's execution log:

- A single 3-language News run (6,000) + 1 Music upload (1,600) = 7,600 units
- This leaves only 2,400 units — not enough for a second News run or more Music tracks
- The Japanese channel is frequently skipped due to quota exhaustion, directly impacting our Japanese-speaking audience

Requested Quota

Metric	Current	Requested
Daily quota	10,000 units	60,000 units

60,000 units/day would allow us to:

- Reliably publish to all 4 channels daily without skipping any language
- Run the News pipeline 2× per day for breaking news coverage (12,000 units)
- Upload 3–5 Music tracks per day (4,800–8,000 units)
- Handle retry scenarios when uploads fail due to network issues
- Accommodate planned growth: additional language channels and increased frequency

9. Channel Links

Channel	Type	URL
BTC News (EN)	News & Politics	youtube.com/@bit_news_en
BTC News (KO)	News & Politics	youtube.com/@bit_news_ko
BTC News (JA)	News & Politics	youtube.com/@bit_news_ja
Music	Music	youtube.com/@vex_seo

* Replace with actual channel URLs if different